

SMOTE-DNN Algorithm Minimizing Dataset Sizes in Gas Classification Method Training for Electronic Noses Application

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Abstract

Real-time detection of mixed gases, as applied in electronic noses (e-noses), has garnered significant attention. However, the field of mixed gas detection continues to face challenges, particularly related to the substantial volume of requisite training data and the associated high costs of data acquisition. In this study, we propose a novel approach for gas classification tailored to scenarios characterized by limited sample sizes, introducing the SMOTE-DNN algorithm. Our findings reveal a substantial reduction in the training dataset size, effectively mitigating data collection expenses while achieving comparable performance using only 1% of the original data volume. Notably, this method achieves recognition accuracies of 92.5301% on the public dataset and 96.6667% on the laboratory-collected dataset. This research provides a resource-efficient methodology to enhance the detection efficiency and precision of experimental measurements conducted using e-noses, thereby offering a pragmatic solution to the challenges inherent in real-time mixed gas analysis.

Keywords:

Gas sensor, SMOTE-DNN, E-noses algorithm, limited sample sizes

1. Introduction

Gas sensors have attracted considerable attention in the real-time and swift detection of toxic and flammable gases[1-3], but individual gas sensors may react to interfering gases owing to their high sensitivity, leading to what is known as cross-sensitivity [4-6]. To address this challenge, researchers have been exploring various methods, including the use of composite materials[7-9], ultrasound assistance, and noise analysis[10, 11]. Despite these efforts, the classification detection of mixed gases remains a daunting task.[12]

In recent years, electronic nose (e-nose) gas detection technology has gained widespread adoption across industries [13-15], pharmaceutical treatments [16-19], food testing [20-22], and other sectors due to its remarkable sensitivity and cost-effectiveness. Unlike traditional sensor arrays, e-nose systems enhance feature extraction and pattern recognition capabilities, significantly reducing the impact of cross-sensitivity on gas detection [23-25] and opening up new possibilities for mixed gas classification. Besides, the emerging machine learning methods enable e-noses to identify mixed gases more precisely through qualitative and quantitative analysis. Thus, machine learning algorithms such as Support Vector Machine (SVM) [26], K-Nearest Neighbors (KNN) [27], and Random Forest (RF) [28] are extensively utilized in e-nose systems at present. However, these traditional algorithms also encounter challenges such as increased algorithmic complexity and diminished generalization capability when confronted with intricate gas detection tasks. As a result, there has been a surge of interest in recent years in deep learning algorithms that seamlessly integrate feature extraction and pattern recognition within a unified model [29-31]. However, traditional machine learning and deep learning methodologies typically require large amounts of training data for robust model training, posing significant challenges in data collection, including execution time and economic costs [32-35].

Hence, effectively training algorithmic models to achieve high-precision responses to real data in the context of limited sample sizes has emerged as a critical challenge.

This study endeavors to advance gas classification methodologies by integrating the Synthetic Minority Over-sampling Technique (SMOTE) with Deep Neural Network (DNN) models. This integration gives rise to the innovative SMOTE-DNN algorithm, designed to efficiently and accurately handle e-nose applications, particularly in scenarios with limited sample dataset sizes through strategic data augmentation. The efficacy of the SMOTE-DNN algorithm was rigorously assessed across diverse datasets, demonstrating its robustness, adaptability, and generalizability. Notably, leveraging the SMOTE algorithm for data augmentation resulted in a significant accuracy improvement of 92.5301% on the test set using public datasets, particularly in scenarios with limited samples. Further experiments involving systematic parameter tuning, particularly the Population Size, yielded a remarkable accuracy of 96.6667% on the test set of laboratory-collected datasets, affirming the algorithmic versatility and superior performance. Concurrently, comparative analyses with traditional machine learning algorithms reaffirmed our SMOTE-DNN algorithmic versatility and superior performance across diverse datasets. Tailored for small datasets, this method expedites the development of gas sensors, leading to notable economic benefits. Consequently, this study holds profound significance in bolstering the training efficiency of e-nose algorithms, thus propelling advancements in gas sensing technology.

2. Experimental Section

2.1. Datasets

2.1.1. Public Data

The dataset was collected on an e-nose system over three years and released in the UCI Machine

Learning Repository. This study used a commercially available array of sixteen screen-printed metal-oxide semiconductor gas sensors, produced and marketed by Figaro Inc. The array consisted of sensor devices, with four sensors each of the following types: TGS2600, TGS2602, TGS2610, and TGS2620[36]. The e-nose system contains 16 gas sensors, and 6 kinds of gases were collected, including ammonia, acetaldehyde, acetone, ethylene, ethanol, and toluene. 8 features were extracted from each sensor, making each sample 128-dimensional. The dataset is divided into 10 batches according to the time of collection, as shown in Table 1. The first batch, which comprises 445 samples, was specifically used for validation in the experiments to ensure the consistency of training and testing for open-set recognition. This decision was made due to the small number of toluene samples in batches 2-5.

2.1.2. Laboratory-Collected Data

The laboratory gas sensor data for this study was collected over a span of two months using a custom-developed multimodal sensor system in our lab. We measured the response curves of Co_3O_4 and $x\text{-Ti-Co}_3\text{O}_4$ ($x=1, 3, 5, 7$) sensors at 280°C towards 50 ppm toluene, o-xylene, m-xylene, p-xylene, and ethylbenzene gases. Each gas measurement was conducted individually, with the baseline resistance of the sensors stabilized before testing[37]. Each sensor extracts 30 different features, including peak values, area under the curve, resistance values at various time intervals, and slopes, among others. As a result, each sample is 30-dimensional for each sensor. Before any experiments were conducted, all data underwent a series of preprocessing steps, including noise removal and normalization, to better facilitate subsequent classification and analysis.

2.2. Preparation Procedure of Sample Gases

2.2.1. Public Data

A computer-controlled continuous flow system, equipped with three digital mass flow controllers

(MFCs) from Bronkhorst High-Tech B.V., was employed to deliver gases to the gas sensing chamber. The MFCs have maximum flow rates of 200 ml min⁻¹, 100 ml min⁻¹, and 20 ml min⁻¹, each with an accuracy of ±1%. The system connected to pressurized gas cylinders containing chemical analytes diluted in dry air, while maintaining a constant humidity of 10% relative humidity (measured at 25 ± 1 °C) throughout the experiment.

Target gases (including ammonia, acetaldehyde, acetone, ethylene, ethanol, and toluene) were introduced randomly into the background gas flow. The total flow rate was set to 200 ml min⁻¹ and held constant during measurements. Sensors operated at a fixed temperature of 400 °C, determined by a manufacturer-provided lookup table, using an integrated heater powered by an external DC power source set to 5 V.

2.2.2. Laboratory-Collected Data

A specified amount of toluene liquid was injected into a closed 5-liter gas chamber, and the targeted gas concentration was acquired through the static liquid gas distribution method according to Eq.1[38-40]:

$$V_1 = \frac{V_2 \times c \times M}{22.4 \times d \times \rho} \times 10^{-9} \times \frac{273 + T_R}{273 + T_B} \quad \text{Eq.(1)}$$

where V_1 is the injected volume of toluene liquid (mL), V_2 is the chamber volume (5000 mL), c is the concentration (ppm) of toluene gas in the chamber, M is the relative molecular weight (92.14 g·mol⁻¹), 22.4 is the gas molar volume (L·mol⁻¹) at 273 K, d and ρ represent the volume fraction (0.991) and density (0.87g·mL⁻¹) of toluene liquid, and T_R and T_B are the temperatures (°C) in the measurement atmosphere and chamber, respectively. The indoor clean air was employed as the main air source, and the environmental humidity could be monitored in real-time via a humidity sensor. The toluene liquid was injected into the heating saturator, which was heated to a temperature of 112 °C to

accelerate the evaporation rate. Therefore, the injected toluene liquid was quickly vaporized and dispersed throughout the entire gas chamber with the assistance of the heating saturator and the two fans, the injected toluene liquid evaporated rapidly throughout the entire gas chamber. Additionally, within less than 2 seconds of opening the gas chamber, the target gas could quickly disperse into the surrounding atmosphere, accompanied by fresh air dispersing near the sensing material during the recovery phase.

2.3. Concentration Information of Gases Used

2.3.1. Public Dataset :

The gases used were not 100% pure but diluted to varying concentrations ranging specifically from 10 to 1000 ppm in dry air.

Detailed concentration values for each gas used in the experiments are listed in Table 2, indicating the exact concentrations utilized.

2.3.2. Laboratory-Collected Data :

99.1% pure target gases were used to create the sample gas mixtures. These target gases were then diluted to specific concentrations using a flow-controlled system. The concentrations of each gas in the air used as sample gas are as follows:

- **Toluene:** 50 ppm
- **Ethylbenzene:** 50 ppm
- **o-Xylene:** 50 ppm
- **m-Xylene:** 50 ppm
- **p-Xylene:** 50 ppm

2.4. Platform Setup

Our experimental environment was carefully configured to ensure consistency in model training and

testing. We conducted all our algorithm implementations in PyCharm (Community Edition, Version 2021.2.2) using Python 3.9. Our experimentation took place on a Windows 10 (x64) operating system, with hardware specifications that included an Intel Core i7-10700 CPU, NVIDIA GTX 1660Ti GPU, and 16GB of RAM. To expedite the training process for deep learning tasks, we harnessed the power of GPU parallel computing. Specifically, we leveraged CUDA™ (Compute Unified Device Architecture), an all-encompassing parallel computing architecture developed by NVIDIA. This architecture includes the CUDA instruction set architecture (ISA) and an embedded parallel computing engine within the GPU. For our deep learning endeavors, we opted for PyTorch (Version 1.8.1) with CUDA support as our primary computational tool.

2.5. SMOTE-DNN Algorithm Design

The pivotal strength of the SMOTE-DNN algorithm lies in its robust generalization across diverse datasets. Through meticulous experimentation, we showcased its prowess on a public dataset spanning six gas classes and a laboratory-collected dataset featuring five gas classes. The algorithm consistently exhibited high accuracy during training, validation, and, most importantly, in scenarios where it encountered entirely new datasets. This adaptability underscores its potential deployment in real-world applications across various industries.

SMOTE stands out as one of the most widely used oversampling techniques, particularly in addressing the challenge of imbalanced class distribution within datasets. This technique aims to rectify the disparity between the minority and majority classes by oversampling the minority class through the generation of synthetic instances. The fundamental principle behind SMOTE involves the creation of synthetic minority instances along the line segments connecting actual instances of the minority class and their nearest neighbors within that class. By introducing these synthetic instances

strategically, SMOTE effectively augments the minority class representation in the dataset. These synthetic instances are carefully positioned in proximity to the existing minority class instances, contributing to the establishment of a more comprehensive decision region for the minority class in the feature space. This augmentation serves to address the imbalance in class distribution, thereby enhancing the classification performance specifically for the minority class. The efficacy of the SMOTE algorithm in mitigating challenges arising from class imbalance has been extensively documented in the literature, underscoring its significance in improving the performance of machine learning models, particularly in scenarios where one class is underrepresented.

In the context of this study, the SMOTE algorithm is applied to augment small-sample data by a factor of 100. This substantial augmentation is aimed at creating a more robust and representative dataset, which is particularly crucial when working with limited samples. Additionally, the introduction of the population size metric further refines the augmentation process. The population size metric is utilized to gradient-train the algorithm with respect to the number of populations, introducing a nuanced approach to determining the optimal quantity of augmented gas feature data. The SMOTE structure is shown in Fig.1(a).

The algorithmic process involves several key steps:

(i) Nearest Neighbors Selection: For each minority sample in the dataset, the algorithm identifies its nearest neighbors within the same class.

(ii) Synthetic Instance Generation: Based on the oversampling rate, a certain number of synthetic samples are generated for each minority sample. These synthetic samples are positioned along the line segments connecting the original minority sample to its nearest neighbors.

Population Size Adjustment: The introduction of the population size metric allows for fine-tuning

the augmentation process. This metric is leveraged to gradient-train the algorithm, adjusting the quantity of synthetic instances generated to optimize the representation of the minority class.

By meticulously executing these steps, the SMOTE algorithm not only addresses the imbalanced class distribution but also tailors the augmentation process to the specific characteristics of the dataset, contributing to the creation of a more effective and representative dataset for subsequent machine learning model training. Additionally, the introduction of the population size metric is utilized to gradient-train with respect to the number of populations.

Deep Neural Network (DNN) is a neural network model consisting of multiple layers of neurons (commonly referred to as hidden layers), with each hidden layer containing multiple neurons. Each neuron receives inputs from the neurons in the preceding layer and computes its output through a weighted sum with an activation function. By propagating data through layers and adjusting weights using optimization algorithms such as gradient descent, DNNs can automatically learn higher-level feature representations from data. This enables DNNs to excel in tasks such as image classification, object detection, speech recognition, natural language processing, and more. The typical DNN structure is shown in Fig.1(b).

The DNN employed in this study is designed to classify five different types of gases based on features extracted from a polymorphic sensor system. The architecture of the DNN has been meticulously chosen to ensure it captures the underlying complexities of the data and exhibits robust generalization capabilities to unseen samples.

The architecture of the DNN comprises the following layers:

(i) Input Layer: The number of neurons in the input layer corresponds to the number of features extracted from each sensor.

(ii) Hidden Layer 1: The first hidden layer consists of 64 neurons and utilizes the Rectified Linear Unit (ReLU) activation function.

(iii) Hidden Layer 2: The second hidden layer also comprises 64 neurons but employs a different activation function, such as the hyperbolic tangent (tanh).

(iv) Dropout Layer: A dropout layer is introduced with a dropout rate of 0.5 to mitigate overfitting.

(v) Hidden Layer 3: The third hidden layer consists of 32 neurons, again utilizing the ReLU activation function.

(vi) Output Layer: The number of neurons in the output layer corresponds to the desired number of gas types to be identified. The activation function used is the Softmax function, transforming the network output into probability scores for each category.

The network is trained using the Adam optimizer with a learning rate of 0.001. The loss function employed is categorical cross-entropy, suitable for multi-class classification problems.

3. Results and Discussion

3.1. Parameters and Training

The model is trained using the Adam optimizer with a learning rate of 0.001. A batch size of 32 is used for training, and the model runs for 8 epochs. The loss function used is categorical cross-entropy. During training, 80% of the data is used for training, and 20% is set aside for validation. The model performance is evaluated at the end of each epoch, and early stopping is implemented if no improvement in validation accuracy is observed for 3 consecutive epochs. This DNN architecture was chosen after extensive experimentation and hyperparameter tuning, aiming to provide the best trade-off between model complexity and the risk of overfitting, given the small size and imbalanced nature

of our datasets. By combining the SMOTE algorithm for data augmentation and the DNN model for classification, this study aims to build a robust, scalable, and efficient model for gas classification tasks, mainly when limited by the availability of large labeled datasets.

3.2. Evaluation Index

A confusion matrix is also generated to get a detailed view of the model performed in each class. This matrix provides insights into the true positive, true negative, false positive, and false negative rates, offering a more nuanced evaluation of the model performance across multiple classes. To assess the model generalization capabilities, we test it on entirely new datasets that were not involved in either the training or validation processes. The same performance metrics (accuracy, F1 score, and confusion matrix) are used for this evaluation. By employing these diverse performance metrics, we aim to provide a comprehensive assessment of the model capabilities, thereby ensuring its applicability in real-world scenarios and its potential for deployment in gas sensing applications.

3.3. Public Dataset Experimentation

In the experimentation with the public dataset, an initial subset comprising 30 instances, with each gas category consisting of 5 instances, was randomly selected. Through the application of the SMOTE algorithm, this subset was augmented by a factor of 100.

Allocating 80% for training and the remaining 20% for validation. As shown in Fig 2 (a), following ten epochs of training, both the training and validation sets achieved notable 100% accuracy in Fig 2 (b), accompanied by a reduction in the loss function to below 0.1. Subsequently, to assess the effectiveness of the optimization algorithm, the same subset of 30 instances was employed for training without SMOTE augmentation. As shown in Fig 2 (d), despite undergoing 100 epochs of training, the

accuracy on both the training and validation sets stabilized at 83.3333% and 33.3333%, respectively. Furthermore, in Fig 2 (e), the loss function exhibited a persistent fluctuation above 0.7. These outcomes indicate that in the absence of SMOTE augmentation, the model performance on both the training and validation sets is comparatively subpar. The low accuracy and elevated loss function values suggest that the model struggled to learn and generalize unseen data effectively. In contrast, the utilization of the SMOTE algorithm to augment the dataset enabled the model to better adapt to sample imbalances, resulting in improved generalization capabilities, higher accuracy, and lower loss. This result further affirms the effectiveness of the SMOTE algorithm in addressing challenges posed by small sample sizes and imbalanced datasets.

In the subsequent experiments, the model underwent testing on the remaining 409 instances, which had not been involved in the training process. As shown in Fig 2 (c), the model, previously trained on a dataset augmented by a factor of 100 from 30 instances, exhibited outstanding accuracy on the test set, achieving 92.5301%.

In stark contrast, as shown in Fig 2 (f), the model that did not undergo data augmentation demonstrated significantly lower accuracy on the test set, reaching only 69.49397%. This pronounced contrast underscores the substantial performance improvement facilitated by the SMOTE algorithm. Through the oversampling of the training set with SMOTE, the model was better equipped to learn from the sample distribution, effectively addressing the challenges posed by small sample sizes and imbalanced datasets, thereby enhancing its generalization capabilities to previously unseen data.

Consequently, these experimental results underscore the effectiveness of the SMOTE algorithm in addressing challenges related to small sample sizes and imbalanced datasets, as well as in enhancing model generalization performance. In this specific context, using SMOTE for data augmentation

played a pivotal role in boosting the model performance on the test set.

Optimization with population size aims to determine the appropriate sample size of instances per gas class needed for data augmentation using the SMOTE algorithm. The “Population Size” parameter was introduced to specify the number of synthetic instances generated for each minority class sample in the SMOTE algorithm. We systematically adjusted the “Population Size”, ranging from small to large values, and observed the model accuracy on the test set throughout the experimentation process, the accuracy changes are shown in Fig 3 (a)-(e). This systematic tuning provided a solid experimental foundation for identifying the optimal “Population Size”.

As illustrated in Fig 3 (f), determining the optimal “Population Size” to be 6 does not imply that the highest gas detection accuracy is achieved at this value. It is anticipated that a larger Population Size would invariably lead to higher accuracy. Our investigation focuses on identifying the minimal sample size required to achieve relatively high detection outcomes. When the Population Size reaches 6, the rate of increase in gas detection accuracy begins to decelerate. This observation indicates that, within the context of this experiment, employing a SMOTE strategy that randomly samples six sets of data for each gas category yields commendable performance in gas sensing tasks. The results underscore the significance of meticulous parameter tuning, particularly the adjustment of “Population Size,” in enhancing model performance in gas sensing tasks with limited sample volume. This experimental insight offers valuable guidance for future analogous tasks, presenting a viable approach to achieving robust detection outcomes under the constraints of sample availability.

3.4. Validation on Laboratory-Collected Data

To assess the algorithm’s generalizability, the SMOTE-DNN algorithm was extended to the dataset collected in the laboratory setting. We randomly selected 6 instances per gas class, totaling 30 instances,

and augmented the dataset using the SMOTE technique. The augmented dataset was then partitioned into training (80%) and validation (20%) sets. After 8 epochs of training, the model achieved 100% accuracy on both the training and validation sets. When tested on the remaining 30 instances, which were not part of the training process, the model achieved an accuracy of 96.6667%. Notably, applying the DNN algorithm directly to these 30 instances without SMOTE augmentation resulted in an accuracy of only 26.6667%. This stark contrast underscores the outstanding performance of the SMOTE-DNN algorithm on the laboratory-collected dataset, further validating its effectiveness across diverse datasets and highlighting its robust generalization capability.

To further validate the performance superiority of the proposed classification model, we conducted a comparative analysis against traditional classification models including Support Vector Machine (SVM), Random Forest (RF), Linear Discriminant Analysis (LDA), K-Nearest Neighbors (K-NN), Logistic Regression (LR). Each model was trained and predicted on the same dataset, comprising designated training, validation, and test sets. As depicted in Fig 4 (a-b), experimental data were visualized using PCA, and the accuracy of each model was meticulously evaluated, with the results presented in Fig 4 (c). The experimental results demonstrate that the model proposed in this study achieved the highest classification accuracy, as shown in Fig 4 (d), reaching 96.67%. This finding underscores the model's efficacy and its substantial potential for predictive accuracy compared to conventional counterparts.

4. Conclusions

In summary, the SMOTE-DNN algorithm introduced in this study represents a paradigmatic shift in gas classification methodologies. By integrating SMOTE for data augmentation with DNN models, this innovative approach has demonstrated exceptional efficiency, adaptability, and accuracy,

particularly in scenarios with limited sample sizes. Experimental results from two distinct datasets—a public dataset and a laboratory-collected dataset—underscore the algorithm’s superior performance across training, validation, and testing phases. In the public dataset experiments, the utilization of the SMOTE algorithm for data augmentation achieved a remarkable accuracy of 92.5301% on the test set, especially in scenarios with limited samples. The significant enhancement in accuracy compared to models without SMOTE augmentation highlights the pivotal role of the SMOTE-DNN algorithm in augmenting model performance under small sample conditions. In the experiments with the laboratory-collected dataset, systematic tuning of the “Population Size” parameter led the algorithm to attain a significant accuracy of 96.6667% on the test set. Comparative experiments with traditional machine learning algorithms further confirmed the algorithm’s universal applicability and superior performance across different datasets. An essential contribution of this study is the robust generalization of the algorithm. In both experiments, the SMOTE-DNN algorithm demonstrated excellent performance on unknown data, validating its robustness in handling unseen scenarios. This strong generalization capability supports the algorithm’s practical application across various gas-sensing contexts. In consequence, the SMOTE-DNN algorithm not only effectively addresses challenges in gas sensing but also sets a new trajectory for future research and application in gas classification methodologies. It offers a novel and promising approach characterized by efficiency, adaptability, and high precision.

Supporting Information

The Supporting Information is available free of charge at XXX.

Validation and training set confusion matrix of public and laboratory-collected dataset, loss and accuracy result of public and laboratory-collected dataset, 3D distribution of samples after dimensionality reduction through PCA, feature description (PDF)

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Notes

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Figures

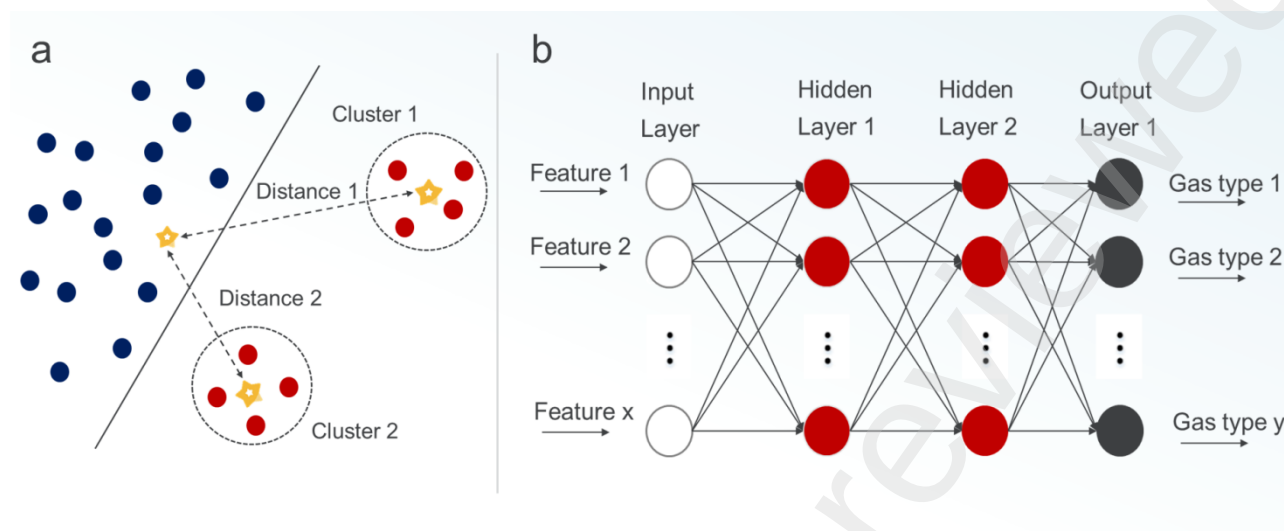


Fig 1. Experimental part schematic diagram. (a) The SMOTE structure. (b) The DNN structure.

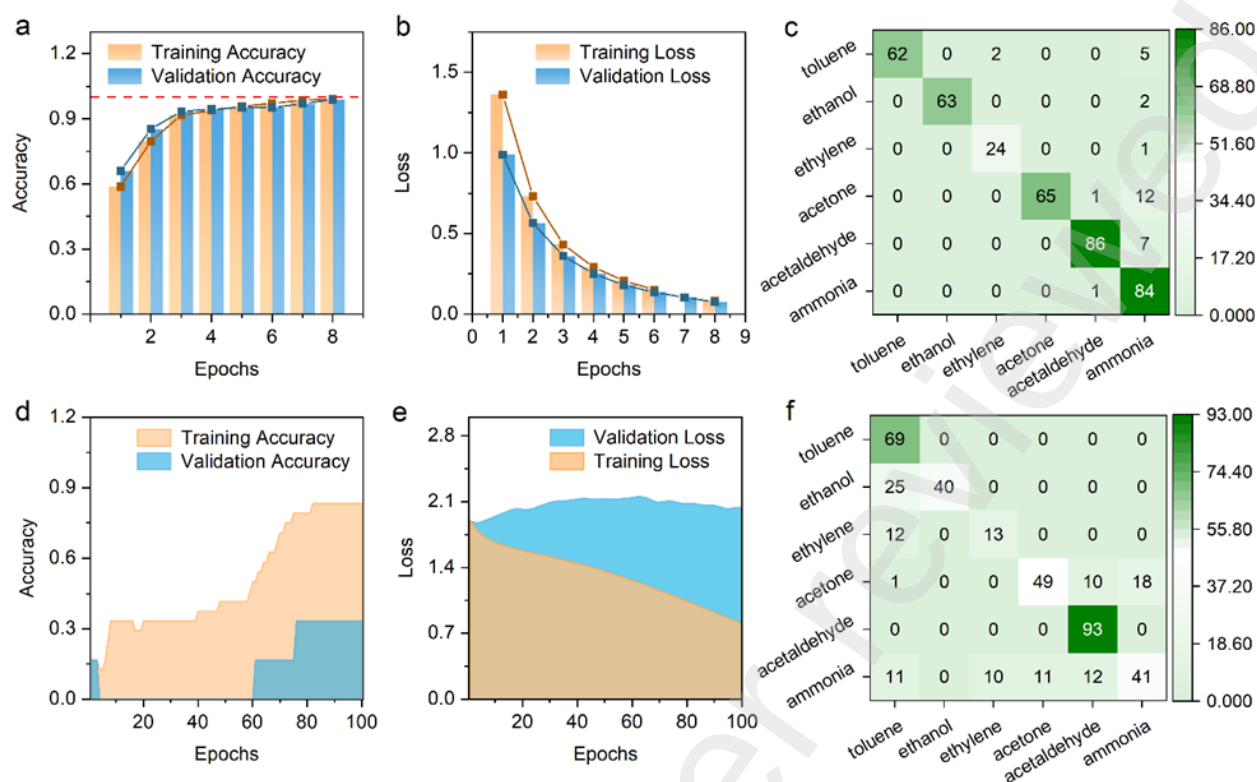


Fig 2. The Results of Public Dataset Experimentation in 30 with SMOTE-DNN. (a) Accuracy result of SMOTE-DNN. (b) Loss result of SMOTE-DNN. (c) The Confusion Matrix of Test set with SMOTE-DNN. (d) Accuracy result of DNN. (e) Loss result of DNN. (f) The Confusion Matrix of Test set without SMOTE-DNN.

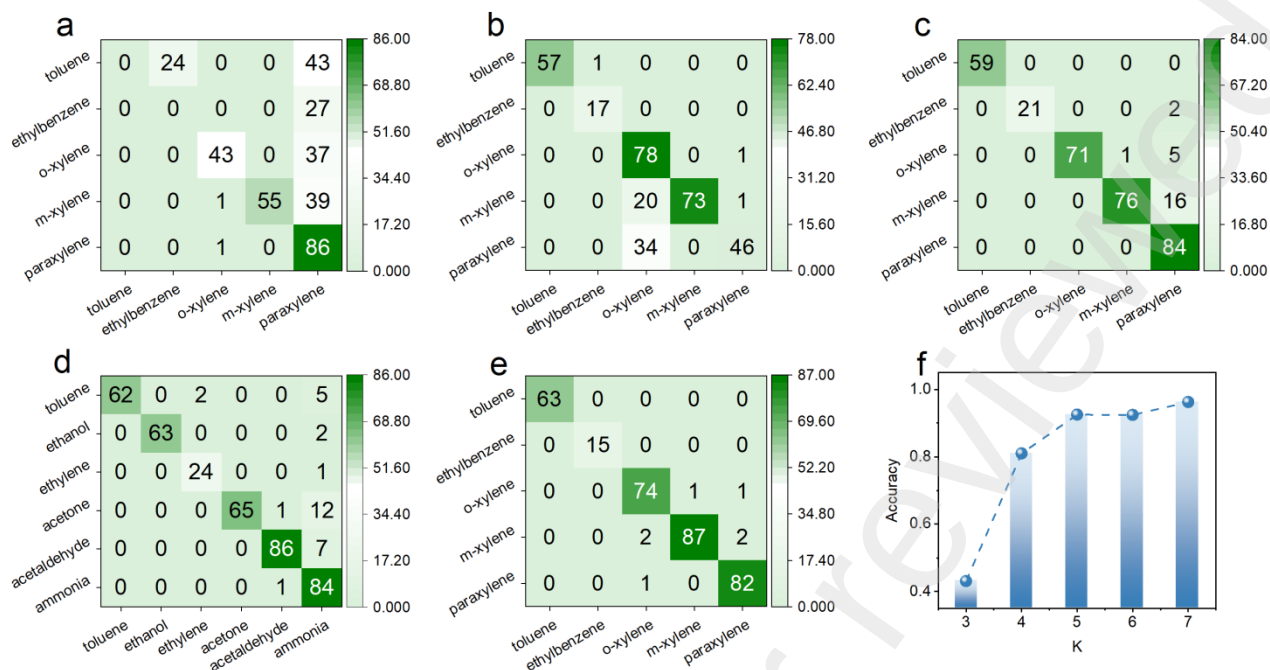


Fig 3. Optimization with Population Size. The test set Confusion Matrix of Public Dataset Experimentation in (a) 18, (b) 24, (c) 30, (d) 36, and (e) 42 with SMOTE-DNN. (f) Accuracy change trend chart with Population Size.

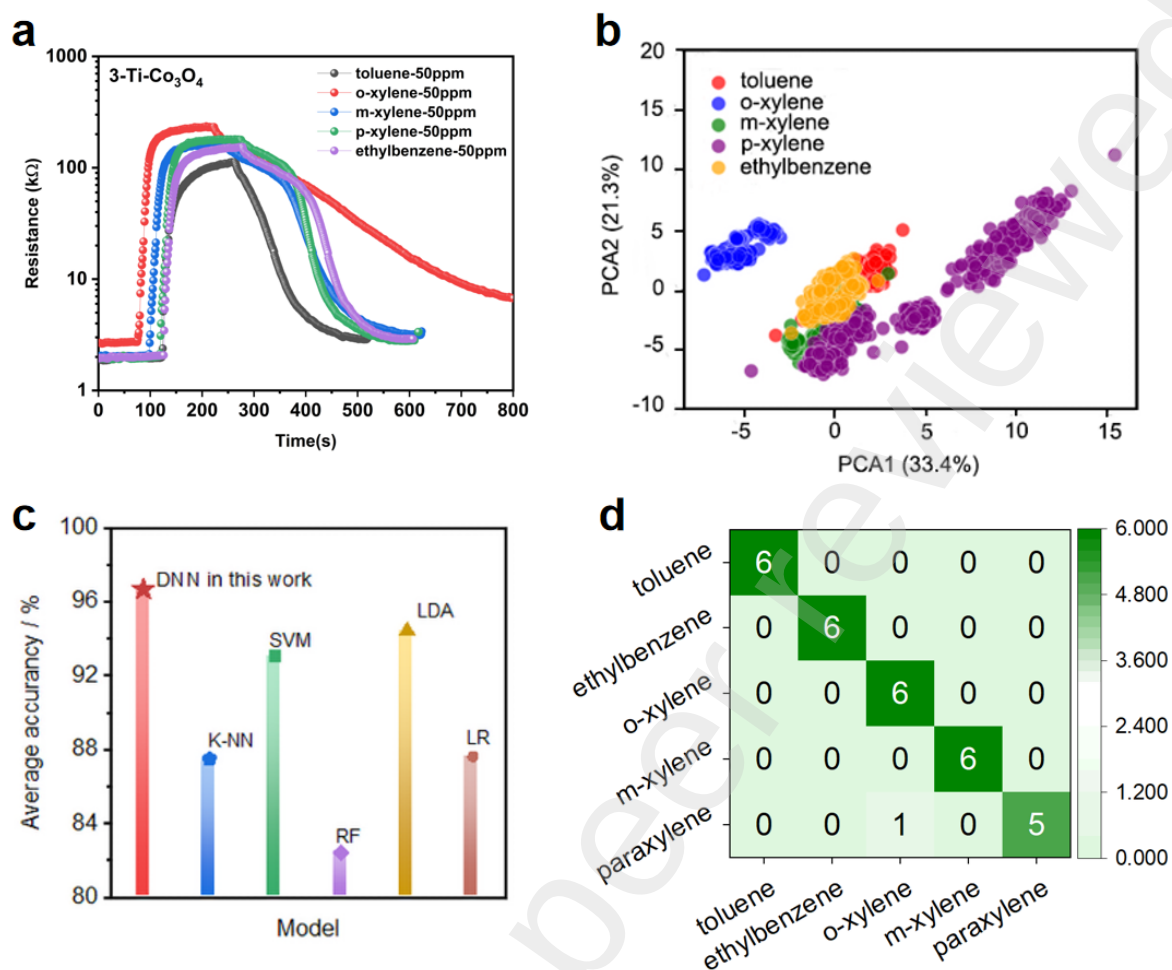


Fig 4. Result analysis. (a) Response–recovery curves of a single 3-Ti-Co₃O₄ sensor for 50 ppm toluene, o-xylene, m-xylene, p-xylene, and ethylbenzene gases in the laboratory. (b) PCA from the extracted features in the response–recovery curves data set. (c) Comparison of classification accuracy with DNN, KNN, SVM, RF, LDA, and LR classifiers. (d) Confusion matrix for the test set using DNN.

Table 1. Details of the public dataset.

Batch	No 1	No 2	No 3	No 4	No 5	No 6	No 7	No 8	No 9	No 10
Acetone	90	164	365	64	28	514	649	30	61	600
Acetaldehyde	98	334	490	43	40	574	662	30	55	600
Ethanol	83	100	216	12	20	110	360	40	100	600
Ethylene	30	109	240	30	46	29	744	33	75	600
Ammonia	70	532	275	12	63	606	630	143	78	600
Toluene	74	5	0	0	0	467	568	18	101	600
Total	445	1244	1586	161	197	2300	3613	294	470	3600

Table 2. Analytes and concentrations in the dataset.

Analytes	Concentrations in ppmv
Ammonia	50, 60, 70, 75, 80, 90, 100, 110, 120, 125, 130, 140, 150, 160, 170, 175, 180, 190, 200, 210, 220, 225, 230, 240, 250, 260, 270, 275, 280, 290, 300, 350, 400, 450, 500, 600, 700, 750, 800, 900, 950, 1000
Acetaldehyde	5, 10, 13, 20, 25, 30, 35, 40, 45, 50, 60, 70, 75, 80, 90, 100, 120, 125, 130, 140, 150, 160, 170, 175, 180, 190, 200, 210, 220, 225, 230, 240, 250, 275, 300, 500
Acetone	12, 25, 38, 50, 60, 62, 70, 75, 80, 88, 90, 100, 110, 120, 125, 130, 140, 150, 170, 175, 180, 190, 200, 210, 220, 225, 230, 240, 250, 260, 270, 275, 280, 290, 300, 350, 400, 450, 500, 1000
Ethylene	10, 20, 25, 30, 35, 40, 50, 60, 70, 75, 90, 100, 110, 120, 125, 130, 140, 150, 160, 170, 175, 180, 190, 200, 210, 220, 225, 230, 240, 250, 275, 300
Ethanol	10, 20, 25, 30, 40, 50, 60, 70, 75, 80, 90, 100, 110, 120, 125, 130, 140, 150, 160, 170, 175, 180, 190, 200, 210, 220, 225, 230, 240,

	250, 275, 500, 600
Toluene	10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100

Graphical Abstract

